Journal of Science and Medicine in Sport xxx (xxxx) xxx

Contents lists available at ScienceDirect

Journal of Science and Medicine in Sport

journal homepage: www.elsevier.com/locate/jsams

Review

How does sleep help recovery from exercise-induced muscle injuries?

Mounir Chennaoui [a,b,*], Théo Vanneau [a,b], Aurélie Trignol [a,b], Pierrick Arnal [c], Danielle Gomez-Merino [a,b], Christophe Baudot [d], Jesus Perez [d], Sebastiano Pochettino [d], Cristiano Eirale [d,e], Hakim Chalabi [d,e,*]

[a] French Armed Forces Biomedical Research Institute (IRBA), France
[b] Paris University, VIgilance FAtigue SOMmeil unit (VIFASOM) EA 7330, France
[c] Dreem, Science Team, New York, USA
[d] Paris Saint Germain FC, France
[e] Aspetar Sports and Orthopedics Hospital, Qatar

ARTICLE INFO

Article history:
Received 3 July 2020
Received in revised form 20 March 2021
Accepted 11 May 2021
Available online xxxx

Keywords:
Exercise injuries
Muscle regeneration
Sleep deprivation
Sleep extension

ABSTRACT

Objectives: Athletes and military personnel may experience sleep disturbances due to conditions of training and competitions or military missions/field operations. The risk of muscle injuries is greater for them when sleep duration decreases, and training load increases simultaneously, which can be exacerbated by fatigue. Accumulating evidence demonstrates that sleep extension improved performance, pain sensitivity and GH/IGF-I anabolic responses, which may be beneficial in accelerating recovery from muscle injuries.
Design & Methods: This narrative review describes the importance of sleep for the recovery/prevention of exercise-induced muscle injuries and provides perspectives on the transferability of currently available scientific evidence to the field.
Results: The first part presents the role of sleep and its interaction with the circadian system for the regulation of hormonal and immune responses, and provides information on sleep in athletes and soldiers and its relationship to injury risk. The second part is an overview of muscle injuries in sport and presents the different phases of muscle regeneration and repair, i.e. degeneration, inflammation, regeneration, remodeling and maturation. Part three provides information on the deleterious effects of sleep deprivation on muscle tissue and biological responses, and on the benefits of sleep interventions. Sleep extension could potentially help and/or prevent recovery from exercise-induced muscle-injuries through increasing local IGF-I and controlling local inflammation.
Conclusions: Although the science of sleep applied to sport is still an emerging field, the current scientific literature shows many potential physiological pathways between sleep and exercise-related muscle injuries. More direct studies are needed to establish clear guidelines for medical personnel and coaches.
© 2021 Sports Medicine Australia. Published by Elsevier Ltd. All rights reserved.

Practical recommendations

Raising athletes' and military soldiers' awareness of the multiple consequences of sleep deprivation has become essential to encourage them to prioritize getting optimal sleep in order to promote recovery processes after muscle injuries, and as well as to prevent injuries. The development of sleep interventions and also diet-based interventions remained important for increasing sleep duration, improving the quality and quantity of sleep, and consequently pain sensitivity and psychological well-being. Specific recommendations to improve the quality of sleep may be to limit exposure to bright light from smartphones or tablets, and to noise in and outside the bedroom, and also to adjust the ambient temperature, and limit fatigue linked to combined stressors. The concept of sleep extension could lead to better regeneration of a muscle injury given the increase in IGF-1 levels and possibly the control of local inflammation. Future studies are necessary in humans and animals to assess the impact of sleep optimized strategies on physiological and clinical mechanisms of recovery from exercise induced muscle injuries and the consecutive athletic performances.

1. Introduction

What is sleep?

Sleep is a basic physiological need and is an important component of human life. Sleep in most individuals occupies between 20% and 40% of the day, and serves multiple restorative functions in the body and the brain through numerous biological processes and systems. Sleep improves memory recall, regulates glucose metabolism, reduces mental fatigue, and plays important role in tissue repair, synaptic homeostasis, and immune-inflammatory responses.[1–3] The restorative function of

* Corresponding authors.
E-mail addresses: mounirchennaoui@gmail.com (M. Chennaoui), hchalabi@psg.fr (H. Chalabi).

https://doi.org/10.1016/j.jsams.2021.05.007
1440-2440/© 2021 Sports Medicine Australia. Published by Elsevier Ltd. All rights reserved.

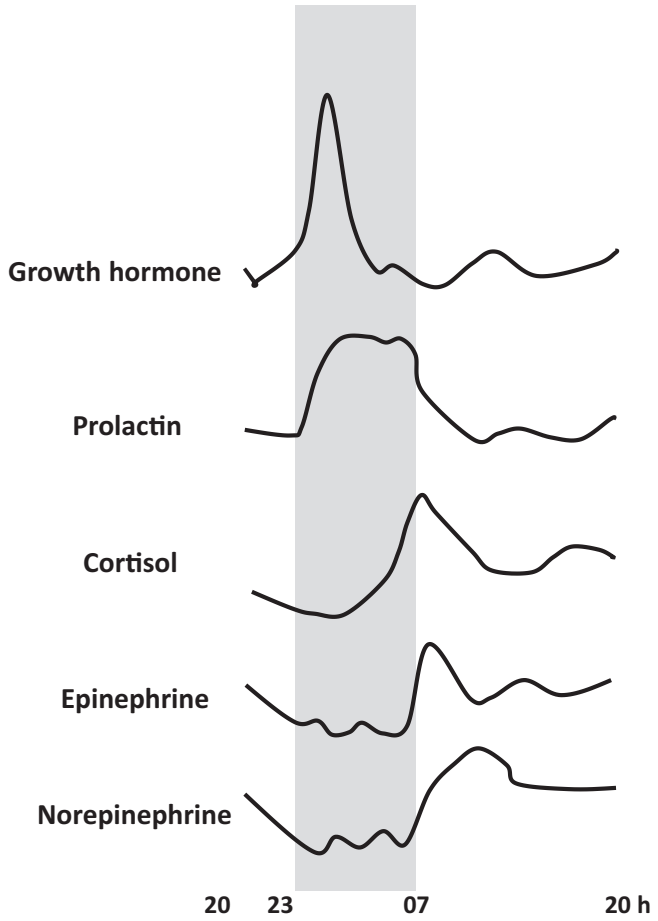

Fig. 1. Effects of sleep and the circadian system on levels of growth hormone (GH), prolactin, cortisol, epinephrine, and norepinephrine in healthy young men during 24 h under conditions of habitual sleep.
Adapted from Lange et al., 2010[61]

sleep also includes the removal of toxic waste byproducts in the brain which have accumulated throughout the day.[1] To promote health, it is recommended that adults sleep at least 7 h per night.

Sleep in humans is divided into two major phases, the first being non-rapid eye movement (NREM) which is predominant during the early part of sleep and decreases in intensity and duration across the sleep period, and the second is rapid eye movement (REM) (also called paradoxical sleep) which becomes more intense and extensive towards the end of the sleep period.[4] The NREM sleep includes three stages, N1, N2, and N3 the deeper or delta wave sleep. Sleep is an important physiologic process controlled in part by an internal circadian clock and in part by a homeostatic mechanism where the pressure for sleep increases in proportion to the duration of prior wakefulness.[2] The circadian system regulates physiological and behavioral processes such as the sleep–wake cycle, locomotor activity, cardiovascular and digestive processes, hormones secretion, body temperature, metabolism, and immunity (Fig. 1).[5]

Sleep and its different stages are characterized by their respective neurochemical milieu of neurotransmitters and hormones and the activity patterns in various endocrine systems. Interaction between sleep and the circadian system regulates hormonal control including melatonin, the hypothalamic-pituitary-adrenal (HPA) axis with cortisol, the hypothalamic–pituitary–thyroid axis, and epinephrine. Moreover, sleep is accompanied by a marked increase in growth hormone (GH), prolactin, and melatonin release, as well as downregulation of the HPA axis and the sympathetic nervous system (SNS).[5,6] Consequently, there is a nocturnal decrease in plasma cortisol, epinephrine and norepinephrine levels. A rapid increase in plasma thyroid stimulating hormone (TSH) levels is also observed in the early evening, beginning 4–5 h before the habitual bedtime. However, the influence of sleep on GH secretion is stronger than the circadian system's influence. A major increase of GH levels occurred soon after sleep onset irrespective of the time of day that sleep takes place, and a surge occurs fairly soon after recommencement of sleep when nocturnal sleep is interrupted. The GH secretion is observed preferentially during slow-wave sleep.

The interaction between sleep and the circadian system also exerts a strong regulatory influence on immune function. During nocturnal sleep, hormones with proinflammatory action and cytokines are synchronized to facilitate the initiation of adaptive immune responses in lymph nodes.[7] Acute and chronic sleep deprivation causes an imbalance in the catabolic/anabolic hormonal ratio and increases markers of systemic inflammation. Increases in the catabolic cortisol, catecholamines and TSH levels have been previously observed and also decreases in the anabolic hormones including testosterone, GH and the GH-related growth factor IGF-I.[3,5,6] In support of these observations on the importance of sleep, numerous studies indicated that a sleep debt may have adverse consequences on cognitive capacities, vigilance, fatigue, mood disorders, and stress, even in young healthy individuals, through perturbations of endocrine secretions and/or sub-clinical inflammatory responses.[3,8]

Exercise and sleep

Sleep and exercise influence each other through complex and reciprocal interactions, including multiple physiological and psychological pathways. It is now clear that exercise represents a potential complementary or alternative approach to improving adult sleep. A systematic review of previous meta-analyses suggests that exercise is associated with significant improvements in the apnea-hypopnea index, overall sleep quality, subjective sleep, and sleep latency.[9] In this study, the methodological quality ranged from 36% to 64% while quality of evidence was very low. The link between exercise and sleep is in fact subject to multiple moderating factors such as gender, age, fitness level, quality of sleep and exercise characteristics (intensity, duration, time of day, environment, etc.).[10,11] The beneficial impact of exercise on sleep quality likely implicated mood, the interaction of circadian rhythm, and physiological effects on thermoregulation, cardiac and autonomic function, and hormonal and immune responses.[10] The practical recommendations for improving sleep seem on moderate aerobic exercise training.

In contrast, endurance athletes and soldiers with obvious signs of overreaching/overtraining (i.e., high perceived fatigue and decreased performance) reported a progressive decrease in sleep quality indices, slight reductions in sleep quantity, and an increase in injury and upper respiratory tract infections.[12,13]

Sleep in athlete and military populations

Sleep is an important element in optimizing athletic performance and recovery, and the consequences of inadequate sleep are particularly important for elite athletes and tactical military personnel, especially younger ones.[14,15]

The sleep characteristics of athlete and military populations

Previous research has indicated elite athletes have a high prevalence of poor sleep quality, insufficient sleep quantity, and disturbances of the circadian rhythm and sleep patterns due to several competitions.[16,17] A systematic review reported that elite athletes generally show a high overall prevalence of insomnia symptoms characterized by longer sleep latencies, greater sleep fragmentation, non-restorative sleep, and excessive daytime fatigue, despite the limited number of high quality studies.[18] Marked between-sport differences on these symptoms was

observed, with individual sports showing the highest levels of sleep disturbance. The sleep of elite athletes is affected by multiple sport-specific factors (type of sport, training load, training/competition time, chronotype, jet lag, seasonal phases) and environmental factors (sleep environment and electronic multimedia devices), the specific conditions of training and competition being probably the most influential factor leading to sleep variability in elite athletes.[19] Especially, scheduling training sessions or competitions during extremely early or late hours can strongly interfere with circadian and homeostatic rhythms.

Currently, recent studies show that military operational personnel are also often exposed to significant sleep debt associated with climatic, nutritional, psychological, physical and mental stressors.[15] In addition, even when they have opportunities for sleep, several additional environmental disruptors in the military operational context make it difficult to obtain restorative sleep, such as light, noise, temperature, and air pollution.

Sleep and the injury risk in athlete and military populations

When circadian rhythmicity and sleep homeostasis are misaligned, the athletic performance can be affected and injuries can occur.[20] The sleep onset latency and efficiency were altered on the night and in the week before lower limb injury occurrence in a four-month follow-up of an elite male soccer player competing in French League 1 and Union of European Football Associations matches.[21] Nevertheless, most studies on the relationship between reduced sleep duration and injury risk were retrospective and concerned young athletes, and the underlying mechanisms are unclear.[22,23] The sleep duration was also associated with the incidence of musculoskeletal injury in 7576 soldiers (mostly men aged less than 35 years, for 95% and 70% of them) from the US Army Special Operations Command.[24] With respect to performance, impairments in reaction time and cognitive function and high levels of fatigue after sleep deprivation were suggested to contribute to injury risk in adolescent elite athletes.[25] An experimental acute ~30 h of sleep deprivation in team-sport athletes was shown to result in slowed pacing strategies, reduced intermittent-sprint performance, reduced muscle glycogen content, reduced peak voluntary force and activation, and negative perceptual strain.[26] A recent meta-analysis on the sleep of elite athletes, including fifty-four studies with objective sleep outcomes (total sleep time and sleep efficiency), demonstrated impairment of sleep the night of competition compared with previous nights, and that additional factors such as early morning training, increases in training load, travel departure times, jet lag and altitude can additionally impair athletes' sleep.[27] Finally, sleep disturbances were also shown detrimental to the recovery processes particularly in the elite soccer after match, resulting in impaired muscle glycogen repletion and muscle damage repair, altered cognitive function and mental fatigue.[11] All of these results explain why interventions targeting sleep, such as sleep extension and sleep hygiene strategies, have been viewed over the past decade as effective strategies for prevention and/or recovery of muscle injuries and performance in elite and competitive athletes, especially professional soccer teams, triathletes and soldiers.[11,15,28]

2. Muscular injuries and regeneration

Description of sport injuries

Several intrinsic and extrinsic risk factors are involved in the occurrence of muscle injury. Chronological age, previous injuries history, poor flexibility, decreased muscle strength or strength imbalance are intrinsic risk factors involved in lower extremity muscle injuries in soccer players particularly.[11,19] The muscle stiffness/flexibility seems to be an important common functional risk factor. Finally, it is apparent that genetic polymorphisms may play role in the inter-individual variability of the response to exercise-induced muscle damage.[29] It included ACTN3, the gene that encodes the α-actinin-3 protein, TNF-α and IL-6, the pro-inflammatory cytokines, and IGF-2. The genetic may also exert protection against muscle injury, by lowering muscle stiffness particularly, as was found for the estrogen receptor (ESR)-1 gene.[30] Among extrinsic risk factors, inadequate nutrition, poor hydration, equipment use, and influences of training and competition / match-related factors (late match, international competition) on muscle injury occurrence have been extensively analysed, while the contribution of sleep loss has been considered for less than 10 years.[22]

Muscle injuries frequently occur during sport practice and represent 30–50% of all injuries in professional football.[31] Typically, muscle injuries are categorized into direct or indirect depending on their etiology. Direct lesions derive from a direct trauma, which involves the direct action of a reasonable external force. Muscle injuries from indirect trauma are the result of a more complex mechanism that doesn't involve an external force but are the result of a more complex mechanism of forces intrinsic to the muscle itself. However, the biological repair phases of direct and indirect injuries do not differ substantially.

There is a certain lexical confusion in the literature regarding the classification of indirect muscle injuries. In fact, different classifications have been presented, generating a substantial interpretative confusion.[32] Considering how the classification may impact the rehabilitation, it is of utmost importance to standardize definitions and methodologies. Classifications are based on etiological, clinical or radiological criteria. In general, grade 0 lesions don't imply an alteration of the muscle anatomy. Fibers lesions are present in grade I, II and III lesions, which are progressively more extensive, with the third grade providing a functional impairment of the muscle itself. There is evidence that the classification can have a prognostic validity for return to play.[33] In general, muscle injuries imply an absence that may vary from few days in grade 0 to few months (3–5) in grade 3.

The diagnosis of a muscle injury is essentially clinical but often is supported by radiological exams. Imaging plays an essential part in the examination of muscle injuries due to the ability to identify and follow up lesions effectively. Although ultrasound has been utilized for approximately 35 years and despite the increasingly frequent use of MRI in muscle lesions, there has been a new activity in ultrasound for some reasons. In fact, if MRI is considered the gold standard, ultrasound can produce similar results when focusing on muscle injuries. Moreover, it is dynamic, very helpful characteristic when evaluating tears. The correlation between ultrasound and MRI with return to play is still debated.[34]

The immediate treatment of muscle injuries involves immobilization and RICE (Rest, Ice, Compression, Elevation) protocol. Immobilization is required to protect the new granulation tissue and can be achieved simply by applying rigid adhesive taping. The use of crutches may also be recommended in cases of important lesions. After the first few days, early mobilization should start for a better capillary development, regeneration of muscle fibers, and more parallel orientation of the regenerating myofibers in the injured area.[35] The rationale behind the RICE treatment is mainly to reduce hemorrhage. The use of NSAIDs is controversial. Their short-term use reduces pain and the time to return to full activity. However, they may possibly predispose to injury recurrence as a result of pain masking and, on a pure biological level, some basic science researches proved that NSAIDs may inhibit muscle regeneration in the first stages of healing, which is dependent on the inflammatory process.[36]

Finally, numerous injection protocols have been suggested for the treatment of muscle injuries. Corticosteroid, prolotherapy, classic mesotherapy, mesotherapy with omeophatic products, PRP (platelet rich plasma) and others have been advocated. However, no protocol has robust scientific evidence, although anecdotal clinical observations and expert opinions can be observed on each practice.[37]

Muscle regeneration

Despite muscle injuries are various in localization and pathological mechanisms, the process of muscle regeneration shares some

similarities. Indeed, numerous injury models (myotoxins, chemical or mechanical agents) were used to investigate the main cellular and molecular players involved in skeletal muscle repair.[38,39] In the previous studies, while muscle regeneration has been extensively studied in animal and cell culture models, less data are available in the adult human skeletal muscle. Interestingly, in vivo myogenesis in human has been recently documented in detail using microscopy of single muscle fibers in three dimensions, highlighting the importance of the basement membrane in the process of regeneration.[40]

The healing capacity of skeletal muscle tissue firstly and mainly relies on resident muscle stem cells (MuSC) termed satellite cells. These cells are located in a specific microenvironment between the cell membrane of myofibers and the basement membrane (a thin layer of extracellular matrix). After an injury, a process of muscle repair takes place with interrelated phases: myofiber necrosis and acute inflammation then a restorative phase, and finally a tissue remodeling/maturation. At a cellular level, satellite cells are activated by the muscle lesion and massively proliferated to generate myogenic progenitors. Then, they differentiate and fuse to form new myofibers. A subset of activated progenitors returns to quiescence to replenish the stem cell pool to preserve long term repair abilities. This coordinated myogenesis needs specific structural and trophic cues from the microenvironment provided by both the immune response and other resident cells.

The immune response is complex. The neutrophils are the first cells invading the injured area producing cytokines such as Interferon γ (IFNγ), Interleukin-1β (IL-1β) and Tumor Necrosis Factor-α (TNF-α). They provide a sterile inflammatory environment to recruit additional immune cells such as circulating monocytes which differentiate into macrophages, key modulators of myogenesis process. Initially, macrophages hold a proinflammatory M1 profile producing cytokines such as IL-6, IL-1β and TNFα in order to stimulate MuSC activation and proliferation. After the phagocytosis of necrotic cells and metabolic modifications, these M1 macrophages skew to a pro-regenerative M2 profile and participate to stimulation of muscle progenitor differentiation and fusion by secreting other molecules such as growth differentiation factor (GDF)-3 while they continue to produce IGF-I and low level of TNF-α. In fact, the M1/M2 macrophages shift and control of the macrophages phenotype seem to be a critical part of the muscle regeneration. Indeed, high concentrations of TNF-α and IL-6 signaling via the transcriptional regulator STAT3 and the classical NF-kB pathway stimulate myoblast proliferation but inhibit subsequent differentiation through the degradation and destabilization of the myogenic regulatory factors MyoD and myogenin.[41] In contrast, low levels of IL-6 and TNF-α are necessary to facilitate later stages of myogenesis by stimulating myoblast differentiation and fusion.

Other infiltrated cells such as eosinophils and regulatory T cells also participate to myogenesis. Fibro–adipogenic progenitors (FAPs) are a muscle interstitial mesenchymal cell population, which secrete signaling molecules acting on myogenic cells such as IGF-I and IL-6. They also produce an optimal acting by its composition and stiffness or by modifying activities of several growth factors. The number of these cells is regulated via apoptosis by TNF-α from inflammatory macrophages whereas during the regenerative phase, macrophages stimulate their extracellular matrix secretion via TGFβ. MuSC and adjacent vessels display reciprocal interactions to repair in a coordinated manner while endothelial cells promote muscle proliferation then their differentiation via growth factors such as IGF-I and hepatocyte growth factor (HGF). *Re*-innervation is also essential since nerves are damaged during muscle injury despite a majority of mature motoneurons survives at distal axotomy, in close contact with Schwann cells.

The resolution of inflammation is an active process based on different soluble mediators including cytokines and growth factors mainly IGF-I produced by multiple cell types at a local level as well as acting by a systemic action. Despite IGF-I is principally synthesized in the liver, it is also produced locally in skeletal muscle tissue. During muscle repair, IGF-1 is produced all during the process by multiple cells (myogenic, immune and stromal cells) and under several IGF-1 isoforms that could be acting at specific myogenesis stage, both on proliferation and differentiation of MuSC.[42]

Finally, depending on the type of injury, muscle regeneration could be more or less effective due to alterations in the extracellular matrix (ECM) structure, the inflammatory response, the revascularization and reinnervation. One or multiple modifications in this complex temporal and spatial process lead to fat and fibrotic tissue deposition which are observed during aging and pathologic processes. Excess of ECM, caused pro-fibrotic molecules such TGFβ and myostatin, will lead to a gradual impairment of muscle function.

3. Sleep time and muscle regeneration: what is known from sleep deprivation and hypothetic effect of sleep extension

Muscle cell proliferation, differentiation, and growth following injury are controlled by a vast number of soluble factors, including several cytokines and growth factors and hormones.[42] The skeletal muscle, like virtually all cells in the body, has circadian rhythms, and recent studies have begun to demonstrate that disruptions in circadian rhythms (as sometimes encountered by elite and competitive athletes) can be detrimental to skeletal muscle health.[43] Thus, a key clock regulator, BMAL1, has been shown to play a critical role in regulating muscle stem cell expansion to impact muscle regeneration.[44] The repair of exercise-induced muscle injuries will need a clean and fast response from biological factors, for elite and military tactical athletes particularly. Since sleep and circadian systems act as important regulators of the hormonal and immune systems, sleep time may play a crucial role in muscle repair.

Several rodent and human studies have described how acute total sleep deprivation or chronic sleep restriction induced a catabolic hormonal signature (i.e. blood increases of cortisol levels and decreases of the anabolic testosterone and IGF-I) with low-grade inflammation (i.e. mainly increases of TNF-α), vascular damages and endothelial dysfunction.[3,45] In the athlete, sleep deprivation increases basal and post-exercise serum levels of skeletal muscle enzymes or proteins which are accurate indicators of muscle damage, such as creatine kinase (CK), aspartate aminotransferase (AST), alanine aminotransferase (ALT), C-reactive protein (CRP) and myoglobin.[46] In a mouse model, this catabolic environment is associated with muscle impairment (i.e. increase of muscle TNF-α levels),[47] which may slow the regeneration process after muscle injury. Indeed, the catabolic hormonal and inflammatory responses induced by sleep deprivation may result in a highly proteolytic environment characterized by decreased protein synthesis and increased degradation which contribute to an increased risk of muscle injury.[48]

Even if we lack scientific evidence on the relationship between sleep and recovery of muscle injury, it has been interestingly demonstrated in a rodent model that muscle regeneration after experimental injury (cryolesion) is impaired after 96 h of sleep deprivation with reduced muscle IGF-1 and plasma testosterone levels, and that 96 h of sleep recovery is not sufficient to normalize muscle regeneration or IGF-1 levels.[49] In humans, the potential value of sleep interventions to promote an anabolic hormonal profile has been demonstrated by sleep extension. Increasing daily total sleep time by less than 1 h during seven days increases circulating concentration of IGF-1 before and during total sleep deprivation in young healthy men.[50] In the rodent, fourteen days of sleep extension increase circulating IGF-I levels, interestingly coupled with increases in the skeletal muscle and frontal cortex tissues.[51] In comparison, pharmacological interventions using exogenous administration of IGF-I have been one way of showing that increasing IGF-I levels may have favorable effects on muscle (cardiac) damage induced by sleep deprivation.[52] After five days of sleep deprivation, myocardial damage occurred accompanied by elevated levels of cardiomyocyte apoptosis and inflammation including IL-6 and IL-1β production, and all symptoms were ameliorated by daily administration

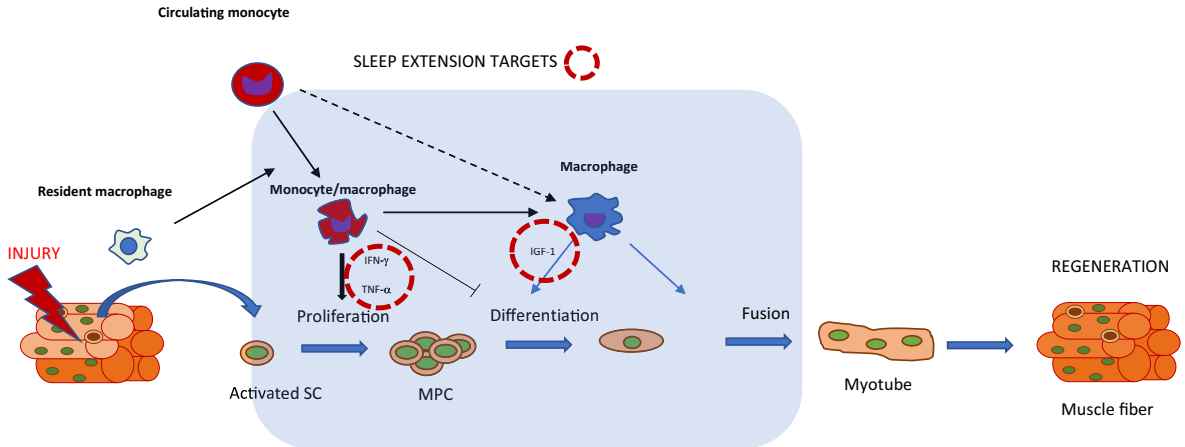

Fig. 2. The mechanistic pathways involved in exercise-induced muscle injuries: possible beneficial effects of sleep extension. Like SCs, resident macrophages are in a quiescent state in healthy muscle but are rapidly activated by exercise-induced injury. Macrophages control proliferation, differentiation and maturation of myocytes during regeneration. Macrophage-derived IGF-1 has been shown a potent coordinator of myogenesis and inflammation in regenerating muscle.[42] We previously evidenced that sleep extension increases blood and skeletal muscle levels of IGF-1,[51] the origin being possibly from resident macrophages. In addition, daily administration of IGF-1 reduced pro-inflammatory cytokines expression in injured myocardium tissue after sleep deprivation.[52] We hypothesized that sleep extension may have beneficial effects on muscle regeneration after injury through IGF-1 and the control of local inflammation. SC: satellite cell, MPC: myogenic precursor cell.
Adapted from Oishi and Manabe, 2018[62]

of IGF-1. Authors demonstrated that the global protective effect of IGF-1 implicated the PI3K/Akt pathway, which enhances protein synthesis when activated.

With respect to the effects of sleep interventions on the inflammation induced by various environmental stressors (i.e., sleep deprivation, muscle injuries after exercise), there is no data in healthy humans. A recent review elegantly demonstrated that high concentrations of TNF-α and IL-6 signaling's stimulate myoblast proliferation but inhibit subsequent differentiation through the degradation and destabilization of the myogenic regulatory factors MyoD and myogenin, while low concentrations are necessary to facilitate later stages of myogenesis by stimulating myoblast differentiation and fusion.[41] Perandini et al.[53] added the interest of chronic physical exercise in restoring the M1 and M2 macrophage balance and consequently, the satellite cell niche to improve skeletal muscle regeneration after injury. We hypothesized that sleep extension may have beneficial effects on muscle regeneration after injury comparable to those of exercise. Future studies on rodents are needed to delineate for example if sleep extension before or after muscle injury may improve the different phases of muscle regeneration and repair. Fig. 2.

Finally, it is interesting to note that several studies have highlighted the beneficial effects of sleep extension or napping on pain sensitivity,[54,55] which may be necessary for recovery from exercise-induced muscle injuries. There is a reciprocal relationship between sleep and pain, with pain disrupting sleep and sleep deprivation or disturbance increasing pain.[56] We can assume that this vicious circle can thus delay the repair of the damaged muscle by limiting early muscle remobilization and preventing good quality of sleep. Pain reduction through sleep interventions and their additional beneficial effects (on alertness, motivation, mood)[57–60] may help elite and military tactical athletes to be diligent in remobilization and medical treatment after muscle injuries and thus indirectly serve the recovery.

Funding information

This work did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of interest

No conflicts of interest declared.

Confirmation of ethical compliance

Not applicable.

Consent for publication

Not applicable.

Authors' contributions

All authors were involved in writing, reviewing and editing the manuscript.

References

1. Eugene AR, Masiak J. The neuroprotective aspects of sleep. *Medtube Sci* 2015;3(1):35-40.
2. Léger D, Debellemaniere E, Rabat A et al. Slow-wave sleep: from the cell to the clinic. *Sleep Med Rev* 2018;41:113-132. doi:10.1016/j.smrv.2018.01.008.
3. Chennaoui M, Léger D, Gomez-Merino D. Sleep and the GH/IGF-1 Axis: consequences and countermeasures of sleep loss/disorders. *Sleep Med Rev* 2020;49:101223. doi:10.1016/j.smrv.2019.101223. Epub 2019 Nov 1.
4. Carskadon MA, Dement WC. *Monitoring and staging human sleep, In: Kryger MH, Roth T, Dement WC, eds. Principles and practice of sleep medicine*, 5th edition., St. Louis, Elsevier Saunders, 2011. p. 16-26.
5. Morris CJ, Daniel Aeschbach A, Scheer FAJL. Circadian system, sleep and endocrinology. *Mol Cell Endocrinol* 2012;349(1):91-104. doi:10.1016/j.mce.2011.09.003. Epub 2011 Sep 10.
6. Steiger A. *Endocrine and metabolic changes during sleep, In: Montagna P, Chokroverty S, eds. Handbook of clinical neurology vol. 98 (3rd series). Sleep Disorders, Part 1*, 2011.
7. Besedovsky L, Lange T, Haack M. The sleep-immune crosstalk in health and disease. *Physiol Rev* 2019;99(3):1325-1380. doi:10.1152/physrev.00010.2018.
8. Irwin MR, Opp MR. Sleep health: reciprocal regulation of sleep and innate immunity. *Neuropsychopharmacology* 2017;42(1):129-155. doi:10.1038/npp.2016.148. Epub 2016 Aug 11.
9. Kelley GA, Kelley KS. Exercise and sleep: a systematic review of previous meta-analyses. *J Evid Based Med* 2017;10(1):26-36. doi:10.1111/jebm.12236.
10. Chennaoui M, Arnal PJ, Sauvet F et al. Sleep and exercise: a reciprocal issue? *Sleep Med Rev* 2015;20:59-72. doi:10.1016/j.smrv.2014.06.008. Epub 2014 Jun 30.
11. Nédélec M, Halson S, Abaidia AE et al. Stress, sleep and recovery in elite soccer: a critical review of the literature. *Sports Med* 2015;45(10):1387-1400. doi:10.1007/s40279-015-0358-z.
12. Lastella L, Vincent GE, Duffield R et al. Can sleep be used as an Indicator of overreaching and overtraining in athletes? *Front Physiol* 2018;9:436. doi:10.3389/fphys.2018.00436. eCollection 2018.
13. Vrijkotte S, Bart Roelands B, Nathalie Pattyn N et al. The overtraining syndrome in soldiers: insights from the sports domain. *Mil Med* 2019;184(5–6):e192-e200. doi:10.1093/milmed/usy274.

14. Simpson NS, Gibbs EL, Matheson GO. Optimizing sleep to maximize performance: implications and recommendations for elite athletes. *Scand J Med Sci Sports* 2017;27(3):266-274. doi:10.1111/sms.12703. Epub 2016 Jul 1.
15. Good CH, Brager AJ, Capaldi VF et al. Sleep in the United States military. *Neuropsychopharmacol* 2020;45(1):176-191. doi:10.1038/s41386-019-0431-7. Epub 2019 Jun 11.
16. Leeder J, Glaister M, Pizzoferro K et al. Sleep duration and quality in elite athletes measured using wristwatch actigraphy. *J Sports Sci* 2012;30(6):541-545. doi:10.1080/02640414.2012.660188. Epub 2012 Feb 14.
17. Knufinke M, Nieuwenhuys A, Geurts SAE et al. Self-reported sleep quantity, quality and sleep hygiene in elite athletes. *J Sleep Res* 2018;27(1):78-85. doi:10.1111/jsr.12509. Epub 2017 Mar 8.
18. Gupta L, Morgan K, Gilchrist S. Does elite sport degrade sleep quality? A systematic review. *Sports Med* 2017;47(7):1317-1333. doi:10.1007/s40279-016-0650-6.
19. Nedelec M, Aloulou A, Duforez F et al. The variability of sleep among elite athletes. *Sports Med Open* 2018;4(1):34. doi:10.1186/s40798-018-0151-2.
20. Smith RS, Efron B, Mah CD et al. The impact of circadian misalignment on athletic performance in professional football players. *Sleep* 2013;36(12):1999-2001. doi:10.5665/sleep.3248.
21. Nédélec M, Leduc C, Dawson B et al. Case study: sleep and injury in elite soccer-a mixed method approach. *J Strength Cond Res* 2019;33(11):3085-3091. doi:10.1519/JSC.0000000000002251.
22. Milewski MD, Skaggs DL, Bishop GA et al. Chronic lack of sleep is associated with increased sports injuries in adolescent athletes. *J Pediatr Orthop* 2014;34(2):129-133. doi:10.1097/BPO.0000000000000151.
23. Gouttebarge V, Aoki H, Ekstrand J et al. Are severe musculoskeletal injuries associated with symptoms of common mental disorders among male European professional footballers? *Knee Surg Sports Traumatol Arthrosc* 2016;24(12):3934-3942. doi:10.1007/s00167-015-3729-y. Epub 2015 Aug 2.
24. Grier T, Dinkeloo E, Reynolds M et al. Sleep duration and musculoskeletal injury incidence in physically active men and women: a study of U.S. Army special operation forces soldiers. *Sleep Health* 2020. doi:10.1016/j.sleh.2020.01.004. S2352-7218(20)30006-1, Online ahead of print.
25. von Rosen P, Frohm A, Kottorp A et al. Multiple factors explain injury risk in adolescent elite athletes: applying a biopsychosocial perspective. *Scand J Med Sci Sports* 2017;27(12):2059-2069. doi:10.1111/sms.12855. Epub 2017 Mar 29.
26. Skein M, Duffield R, Edge J et al. Intermittent-sprint performance and muscle glycogen after 30 H of sleep deprivation. *Med Sci Sports Exerc* 2011;43(7):1301-1311. doi:10.1249/MSS.0b013e31820abc5a.
27. Roberts SSH, Teo WP, Warmington SA. Effects of training and competition on the sleep of elite athletes: a systematic review and meta-analysis. *Br J Sports Med* 2019;53(8):513-522. doi:10.1136/bjsports-2018-099322. Epub 2018 Sep 14.
28. Hotfiel T, Mayer I, Huettel M et al. Accelerating recovery from exercise-induced muscle injuries in triathletes: considerations for Olympic distance races. *Sports (Basel)* 2019;7(6):143. doi:10.3390/sports7060143.
29. Baumert P, Lake MJ, Stewart CE et al. Genetic variation and exercise-induced muscle damage: implications for athletic performance, injury and ageing. *Eur J Appl Physiol* 2016;116(9):1595-1625. doi:10.1007/s00421-016-3411-1. Epub 2016 Jun 13.
30. Kumagai H, Miyamoto-Mikami E, Hirata K et al. ESR1 rs2234693 polymorphism is associated with muscle injury and muscle stiffness. *Med Sci Sports Exerc* 2019;51(1):19-26. doi:10.1249/MSS.0000000000001750.
31. Bengtsson H, Ekstrand J, Waldén M et al. Muscle injury rate in professional football is higher in matches played within 5 days since the previous match: a 14-year prospective study with more than 130 000 match observations. *Br J Sports Med* 2018;52(17):1116-1122. doi:10.1136/bjsports-2016-097399. Epub 2017 Nov 3.
32. Mueller-Wohlfahrt HW, Haensel L, Mithoefer K et al. Terminology and classification of muscle injuries in sport: the Munich consensus statement. *Br J Sports Med* 2013;47(6):342-350. doi:10.1136/bjsports-2012-091448.
33. Ekstrand J, Askling C, Magnusson H et al. Return to play after thigh muscle injury in elite football players: implementation and validation of the Munich muscle injury classification. *Br J Sports Med* 2013;47:769-774. doi:10.1136/bjsports-2012-092092.
34. van der Made AD, Almusa E, Whiteley R et al. Intramuscular tendon involvement on MRI has limited value for predicting time to return to play following acute hamstring injury. *Br J Sports Med* 2018;52(2):83-88. doi:10.1136/bjsports-2017-097659.
35. Järvinen TAH, Järvinen TLN, Kääriänen M et al. Muscle injury. Biology and treatment. *Am J Sports Med* 2005;33(5):745-764. doi:10.1177/0363546505274714.
36. Mishra DK, Fridén J, Schmitz MC et al. Anti-inflammatory medication after muscle injury: a treatment resulting in short-term improvement but subsequent loss of muscle function. *J Bone Joint Surg* 1995;77A:1510-1519. doi:10.2106/00004623-199510000-00005.
37. Orchard JW, Best TM, Mueller-Wohlfahrt HW et al. The early management of muscle strains in the elite athlete: best practice in a world with a limited evidence basis. *Br J Sports Med* 2008;42:158-159. doi:10.1136/bjsm.2008.046722.
38. Chargé SBP, Rudnicki MA. Cellular and molecular regulation of muscle regeneration. *Physiol Rev* 2004;84(1):209-238. doi:10.1152/physrev.00019.2003.
39. Wosczyna MN, Rando TA. A muscle stem cell support group: coordinated cellular responses in muscle regeneration. *Dev Cell* 2018;46(2):135-143. doi:10.1016/j.devcel.2018.06.018.
40. Mackey AL, Kjaer M. The breaking and making of healthy adult human skeletal muscle in vivo. *Skelet Muscle* 2017;7(1):24. doi:10.1186/s13395-017-0142-x.
41. Howard EE, Pasiakos SM, Blesso CN et al. Divergent roles of inflammation in skeletal muscle recovery from injury. *Front Physiol* 2020;11:87. doi:10.3389/fphys.2020.00087. eCollection 2020.
42. Tidball JG. Regulation of muscle growth and regeneration by the immune system. *Nat Rev Immunol* 2017;17(3):165-178. doi:10.1038/nri.2016.150. Epub 2017 Feb 6.
43. Andrews JL, Zhang X, McCarthy JJ et al. CLOCK and BMAL1 regulate MyoD and are necessary for maintenance of skeletal muscle phenotype and function. *Proc Natl Acad Sci U S A* 2010;107(44):19090-19095. doi:10.1073/pnas.1014523107. Epub 2010 Oct 18.
44. Chatterjee S, Yin H, Nam D et al. Brain and muscle Arnt-like 1 promotes skeletal muscle regeneration through satellite cell expansion. *Exp Cell Res* 2015;331(1):200-210. doi:10.1016/j.yexcr.2014.08.041. Epub 2014 Sep 9.
45. Chennaoui M, Sauvet F, Drogou C et al. Effect of one night of sleep loss on changes in tumor necrosis factor alpha (TNF-α) levels in healthy men. *Cytokine* 2011;56(2):318-324. doi:10.1016/j.cyto.2011.06.002. Epub 2011 Jul 6.
46. Mejri MA, Yousfi N, Hammouda O et al. One night of partial sleep deprivation increased biomarkers of muscle and cardiac injuries during acute intermittent exercise. *J Sports Med Phys Fitness* 2017;57(5):643-651. doi:10.23736/S0022-4707.16.06159-4.
47. Yang DF, Shen YL, Wu C et al. Sleep deprivation reduces the recovery of muscle injury induced by high-intensity exercise in a mouse model. *Life Sci* 2019;235:116835. doi:10.1016/j.lfs.2019.116835. Epub 2019 Sep 4.
48. Carbone JW, McClung JP, Pasiakos SM. Skeletal muscle responses to negative energy balance: effects of dietary protein. *Adv Nutr* 2012;3(2):119-126. doi:10.3945/an.111.001792.
49. Mônico-Neto M, Dáttilo M, Ribeiro DA et al. REM sleep deprivation impairs muscle regeneration in rats. *Growth Factors* 2017;35(1):12-18. doi:10.1080/08977194.2017.1314277.
50. Chennaoui M, Arnal PJ, Drogou C et al. Sleep extension increases IGF-I concentrations before and during sleep deprivation in healthy young men. *Appl Physiol Nutr Metab* 2016;41(9):963-970. doi:10.1139/apnm-2016-0110.
51. Chennaoui M, Arnal PJ, Dorey R et al. Changes of cerebral and/or peripheral adenosine A1 receptor and IGF-I concentrations under extended sleep duration in rats. *Int J Mol Sci* 2017;18(11):1-10. doi:10.3390/ijms18112439.
52. Chen H, Zhao Z, Zheng J et al. The effect of IGF-1 on symptoms of sleep deprivation in a rat model of inflammatory heart disease and metabolic syndrome. *Biochem Biophys Res Commun* 2014;446(4):843-849. doi:10.1016/j.bbrc.2014.02.123.
53. Perandini LA, Chimin P, da Silva LD et al. Chronic inflammation in skeletal muscle impairs satellite cells function during regeneration: can physical exercise restore the satellite cell niche? *FEBS J* 2018;285(11):1973-1984. doi:10.1111/febs.14417. Epub 2018 Mar 8.
54. Faraut B, Léger D, Medkour T et al. Napping reverses increased pain sensitivity due to sleep restriction. *PLoS ONE* 2015;10(2):1-16. doi:10.1371/journal.pone.0117425.
55. Simonelli G, Mantua J, Gad M et al. Sleep extension reduces pain sensitivity. *Sleep Med* 2019;54:172-176. doi:10.1016/j.sleep.2018.10.023.
56. Stroemel-Scheder C, Kundermann B, Lautenbacher S. The effects of recovery sleep on pain perception: a systematic review. *Neurosci Biobehav Rev* 2020;113:408-425. doi:10.1016/j.neubiorev.2020.03.028. Epub 2020 Apr 8.
57. Arnal PJ, Sauvet F, Leger D et al. Benefits of sleep extension on sustained attention and sleep pressure before and during total sleep deprivation and recovery. *Sleep* 2015;38(12):1935-1943. doi:10.5665/sleep.5244.
58. Arnal PJ, Lapole T, Erblang M et al. Sleep extension before sleep loss: effects on performance and neuromuscular function. *Med Sci Sports Exerc* 2016;48(8):1595-1603. doi:10.1249/MSS.0000000000000925.
59. Ritland B, Simonelli M, Gentili G et al. Effects of sleep extension on cognitive/motor performance and motivation in military tactical athletes. *Sleep Med* 2019;58:48-55. doi:10.1016/j.sleep.2019.03.013.
60. Halson SL. Sleep in elite athletes and nutritional interventions to enhance sleep. *Sports Med* 2014;44(Suppl 1):S13-S23. doi:10.1007/s40279-014-0147-0.
61. Lange T, Dimitrov S, Born J. Effects of sleep and circadian rhythm on the human immune system. *Ann N Y Acad Sci* 2010;1193:48-59. doi:10.1111/j.1749-6632.2009.05300.x.
62. Oishi Y, Manabe I. Macrophages in inflammation, repair and regeneration. *Int Immunol* 2018;30(11):511-528. doi:10.1093/intimm/dxy054.